

CSC3109 Machine Learning

AY2025/2026 Trimester 3

Group Project Specifications

1. Objectives

The overarching objective of this group project is to translate deep learning concepts into practice by solving a real-world computer vision problem. Students will apply the techniques studied in lectures to design, train, evaluate, and deploy image-classification models on a curated aerial-imagery dataset. The project develops comprehensive competencies in data processing, model architecture selection, training strategies, performance benchmarking, and decision making, while cultivating a constructive attitude towards large-scale machine learning engineering.

2. Key Concepts

Deep learning · Convolutional neural networks · Transfer learning · Real-world computer vision · Data augmentation · Model benchmarking · Containerised deployment · Technical reporting

3. Project Description

Deep Learning (DL) continues to drive technological advancement across many domains, including remote sensing, urban planning, environmental monitoring, and disaster response. Aerial and satellite imagery analysis has emerged as a critical capability, enabling intelligent systems to classify land-use types, detect infrastructure, and monitor large-scale geographic change automatically.

As a deep learning team, your goal is to build a high-performing image-classification system for aerial scene recognition. You will acquire hands-on experience in data exploration, pre-processing, model design, training, evaluation, and deployment—skills that are in high demand across the AI industry. Modern deep learning frameworks such as PyTorch, TensorFlow, MXNet, or equivalent libraries may be used.

The key objective is to understand advanced deep learning techniques, analyse a real-world dataset, and implement original solutions. Each team will receive a dedicated dataset partition and submit solutions for rigorous performance evaluation. All evaluation metrics must be clearly documented in the final report.

Open-source libraries are permitted; however, teams must register every library used and provide clear justifications. The team must also document their own contributions, enhancements, and customisations. Simply downloading and executing an existing repository without meaningful understanding and modification is not acceptable.

3.1) Dataset and Task. Each team will be assigned a dedicated dataset partition drawn from a 38-category aerial-imagery benchmark. The partition contains exactly 4 visually confusable categories (e.g., different types of roads, residential areas, or vegetation), making fine-grained classification the core challenge. Each category provides 700 training images and 100 held-out validation images. The held-out validation images do not appear in the training split and must be used solely for performance evaluation.

3.2) End-to-End Machine Learning Pipeline. Based on a thorough understanding of the problem, teams must provide: clear motivations and problem framing, exploratory data analysis (EDA), data preparation and augmentation, model training and hyper-parameter tuning, inference on the validation set, and comprehensive performance evaluation using standard metrics (accuracy, precision, recall, F1-score, confusion matrix).

3.3) Deep Learning Approaches. Each team must propose and implement multiple deep learning approaches. The number of distinct approaches investigated must be at least equal to the number of team members. Strengths and weaknesses of each approach must be discussed extensively, and key insights derived from the comparative analysis must be presented in the report.

3.4) Deployment via Containerisation. The final, best-performing model must be packaged and deployed through containerisation (e.g., Docker). The container should expose an inference endpoint capable of accepting an aerial image and returning a predicted category label with confidence scores.

4. Assessment Criteria

The group assignment will be assessed according to the marking scheme in Table 1. Criteria are subject to revision based on progress and circumstances throughout the trimester.

Criteria	Weight
Understanding and Solving a Real-World Challenge • Background and motivations • Problem statement and project objectives	20%

• Survey of existing approaches	
Machine Learning Solutions • Exploratory data analysis • Data preparation and pre-processing • Investigation of deep learning approaches (number of approaches $\geq$ number of team members) • Model training and tuning • Deep learning inference • Performance evaluation • Strengths and weaknesses • Results, insights, and discussions • Deployment through containerisation	60%
Report Quality • Professional writing with clear logic and structure • Reflection on learning and application to the project • Future directions	10%
Video Demonstration • 3–5 minute video demonstrating model performance and key findings	10%

** Group marks may be weighted by peer-review scores where necessary.*

NOTE: Bonus marks will be awarded for exceptional depth of understanding, innovative techniques, and insightful analysis beyond the baseline requirements.

5. Timeline and Deliverables

Final submission deadline: Sunday, 26 July 2026, 23:59 (local time).

Each group submits to their designated xSITE Dropbox. One submission per team. The submission package must contain:

- T<Team Number>.pdf — final report (e.g., T01.pdf)
- T<Team Number>.zip — compressed file with video and supplementary materials (e.g., T01.zip)

The final report should be approximately 30 pages (maximum 50 pages) and must include:

Overall Project Description (3–6 pages)

- Background and motivations
- Problem statement
- Project objectives

- Review of existing approaches

Machine Learning Solutions (15–18 pages)

- Exploratory data analysis
- Data preparation and pre-processing
- Investigation of deep learning approaches
- Model training and tuning
- Deep learning inference
- Performance evaluation
- Strengths and weaknesses
- Containerisation
- Results, insights, and discussions

Individual Contributions (1–2 pages)

- Clear and concise description of each team member's contribution

Reflection (2–3 pages)

- What you have learnt and how you applied it to the project

Future Work (1–2 pages)

- Proposed future directions and improvements

6. Teams

Students will be assigned to teams of 5 members in Week 3. Each team will receive a unique dataset partition (4 confusable aerial-image categories) and must develop their own machine learning solution independently from week 4 onward.

7. Late Submission

A penalty of 20% per day will be imposed for late submissions unless an extension has been approved by the lecturers prior to the deadline. Extension requests are considered on a case-by-case basis. Submissions received more than 4 days after the deadline will not be accepted and will receive zero marks.

8. Academic Integrity and Plagiarism

All submitted work must be the team's own. Copying code, models, or written content from another person or team without proper attribution is strictly prohibited. It is each student's responsibility to ensure that their solutions are original and that their work is not made

accessible to others for the purpose of copying. Where plagiarism is detected, all parties involved will receive zero marks for the entire project.